

TIC 403017570

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 0.828992 d
Epoch (t0) 3069.4201 BTJD
Duration 0.1700 d
Depth 0.433 % (4,327 ppm)
SNR 1074.8

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 57 (gate ≥ 3)
Robust depth ratio 1.08 (gate ≥ 0.50)
Robust depth sigma 80.8 (gate ≥ 7.0)
Odd/even z-score 2.52 (consistent)
Secondary eclipse not detected

CATALOG CROSS-MATCH (novelty)

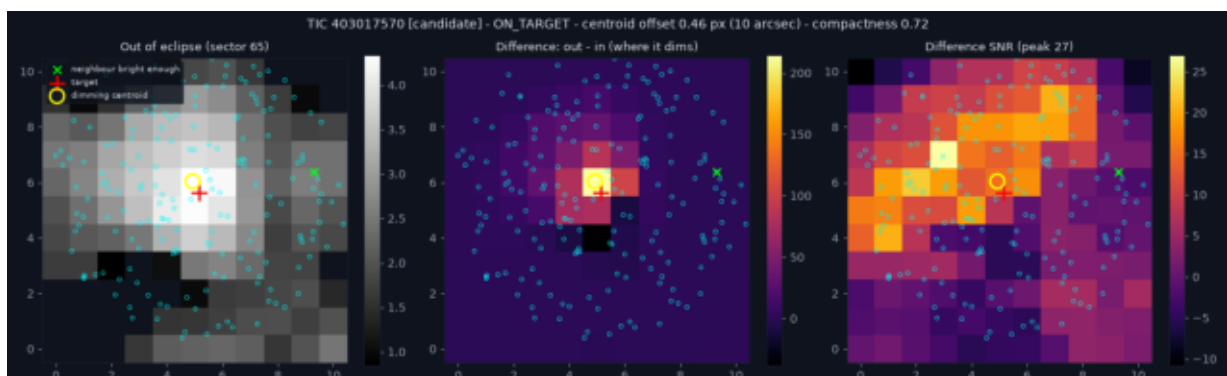
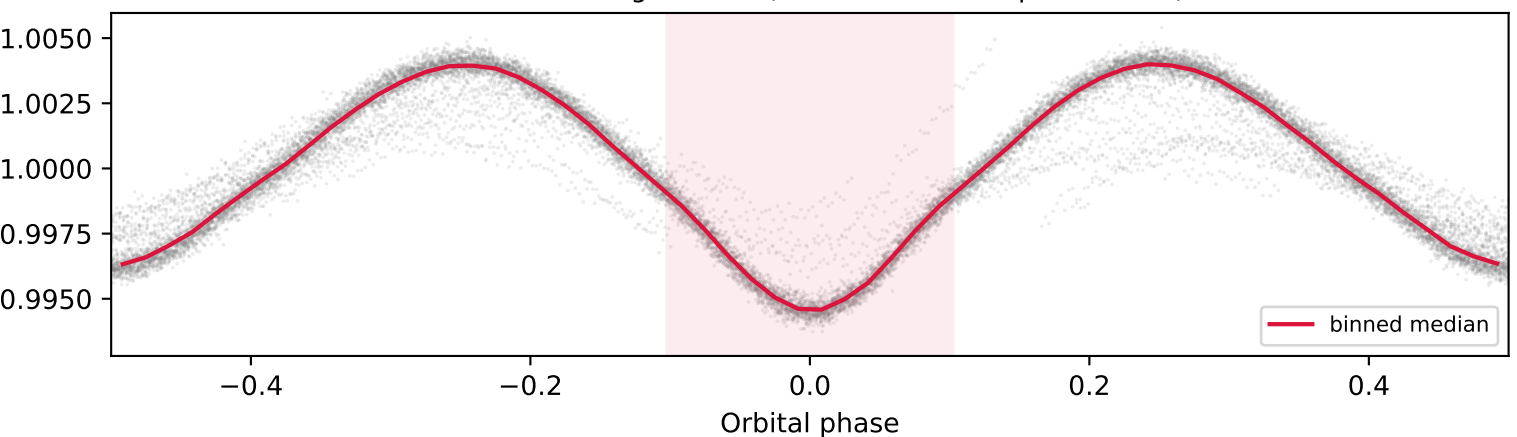
VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 65 light-curve dip in this sector: 220.5 sigma t0 refined by 0.0050 phase
Difference-image peak SNR 27.03 (gate ≥ 5.0) Compactness 0.72 (gate ≥ 0.50)
Dimming centroid offset 0.46 px (10 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 403017574
4.39 px away, dTmag +4.43 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.